Electronic supplementary material

A reproducible benchmark for transformation uncertainty in aquifer-test parameter inference

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Supplementary tables

Table S1: Uncertainty source matrix for separation of field response diagnostics, parameter compensation, and unresolved uncertainty sources.

Source	Evidence and current limitation
Inherent variability	Field COV and variance after removing the trend diagnose variability in inferred hydraulic properties; independent heterogeneity remains a separate evidence need.
Measurement error	The Lovelock detection threshold is included; the Massachusetts site instrument budget remains unavailable.
Pumping schedule and preprocessing	Massachusetts retained constant rate analysis and Lovelock pumping/recovery superposition are reported separately.
Transformation uncertainty	Time and well indexed log model factor dispersion, model factor distributions, diagnostics by time window, and consistency between the two field cases quantify interpretation related response uncertainty.
Model form	AIC/BIC, model definitions, leakage support, and boundary diagnostics compare implemented interpretation models and expose unimplemented model limits.
Statistical and identifiability	Local MAP correlations, boundary flags, leakage support, and field parameter COV diagnose finite sample and tradeoff effects.
Support and decision scale	Spatial transfer, comparison between the two field cases, Bayesian model averaging, robust hydraulic control capacity factors, and response time consequences connect parameter estimation to groundwater use.

Table S2: Parallel headline results for the Massachusetts and Lovelock Valley field cases across the model sequence Theis confined / Hantush–Jacob leaky / lagging Darcy without leakage / lagging Darcy with leakage.

Evidence item	Massachusetts field case	Lovelock Valley field case
Setting and support	Fractured bedrock; 5.8–36.6 m radial support	Basin fill aquifer; 0.6–9.6 mi radial support
Wells / response records	12 / 420	7 / 520
Observation design	Retained constant rate drawdown subset	Pumping and recovery superposition with 0.05 ft detection offset
Pooled log model factor SD_ϵ	0.166 / 0.164 / 0.062 / 0.035	0.246 / 0.294 / 0.174 / 0.174
BIC preferred wells	2 / 2 / 3 / 5	2 / 0 / 5 / 0
Transmissivity COV	0.58 / 0.78 / 0.46 / 0.81	3.40 / 3.70 / 1.83 / 4.77
Storage COV	2.05 / 1.71 / 1.74 / 1.54	0.89 / 0.86 / 0.95 / 10.21
Support for models that include leakage	Lagging leaky BIC preferred in 5 wells, with spatially selective leakage support	BIC preference concentrated in models without leakage
Scope of validation	Spatial transfer diagnostic available for the small well network	Field consistency check available for the processed regional data set

Table S3: Model-averaged management diagnostic for aquifer-test transformation uncertainty across the four formally fitted analytical interpretations, Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage. Rows without a strict BIC prefix use variance-window BMA weights.

Decision diagnostic	Massachusetts field case	Lovelock Valley field case
Mean model weights	0.19 / 0.22 / 0.26 / 0.32	0.25 / 0.15 / 0.34 / 0.26
Selected model counts	2 / 2 / 3 / 5	2 / 0 / 5 / 0
Strict BIC median capacity factor	1.00	1.00
Median capacity factor	1.60	1.18
95th percentile capacity factor	2.73	4.52
Mean capacity shortfall probability	0.38	0.21
Median response factor	1.85	4.05
95th percentile response factor	3.32	8.79
Mean earlier response probability	0.41	0.23
Median diffusivity spread	1.90	3.91

Table S4: Groundwater report endpoint for aquifer-test transformation uncertainty via the evidence sequence from measured response to numerical benchmark support, field applicability, and decision consequence.

Reporting item	Required statement before pumping-test-derived hydraulic properties are used
Measured response and support	State the pumping representation, observation geometry, detection threshold or offset, and retained well records so that the transformed response and spatial support are explicit.
Interpretation model	Identify whether T and S come from Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, lagging Darcy with leakage, or another stated interpretation.
Response log model factor	Report observed-predicted log response diagnostics and pooled SD_ϵ for each interpretation to make response-level transformation error visible.
Complexity and parameter spread	Report information criteria, leakage or boundary support, fitted parameter COV, and identifiability flags before interpreting apparent parameters as hydrogeologic evidence.
Numerical model factor support	Use screened benchmark model factor distributions for matching scenario classes, pathways, and observation supports.
Field applicability check	State the field-to-benchmark class match after response, residual, complexity, and parameter-spread diagnostics so the scenario-conditioned COV values are tied to their support.
Decision consequence	Propagate the supported interpretation set to capacity, diffusivity, and response-time diagnostics so later groundwater control or management calculations know what uncertainty they inherit.

Supplementary figures

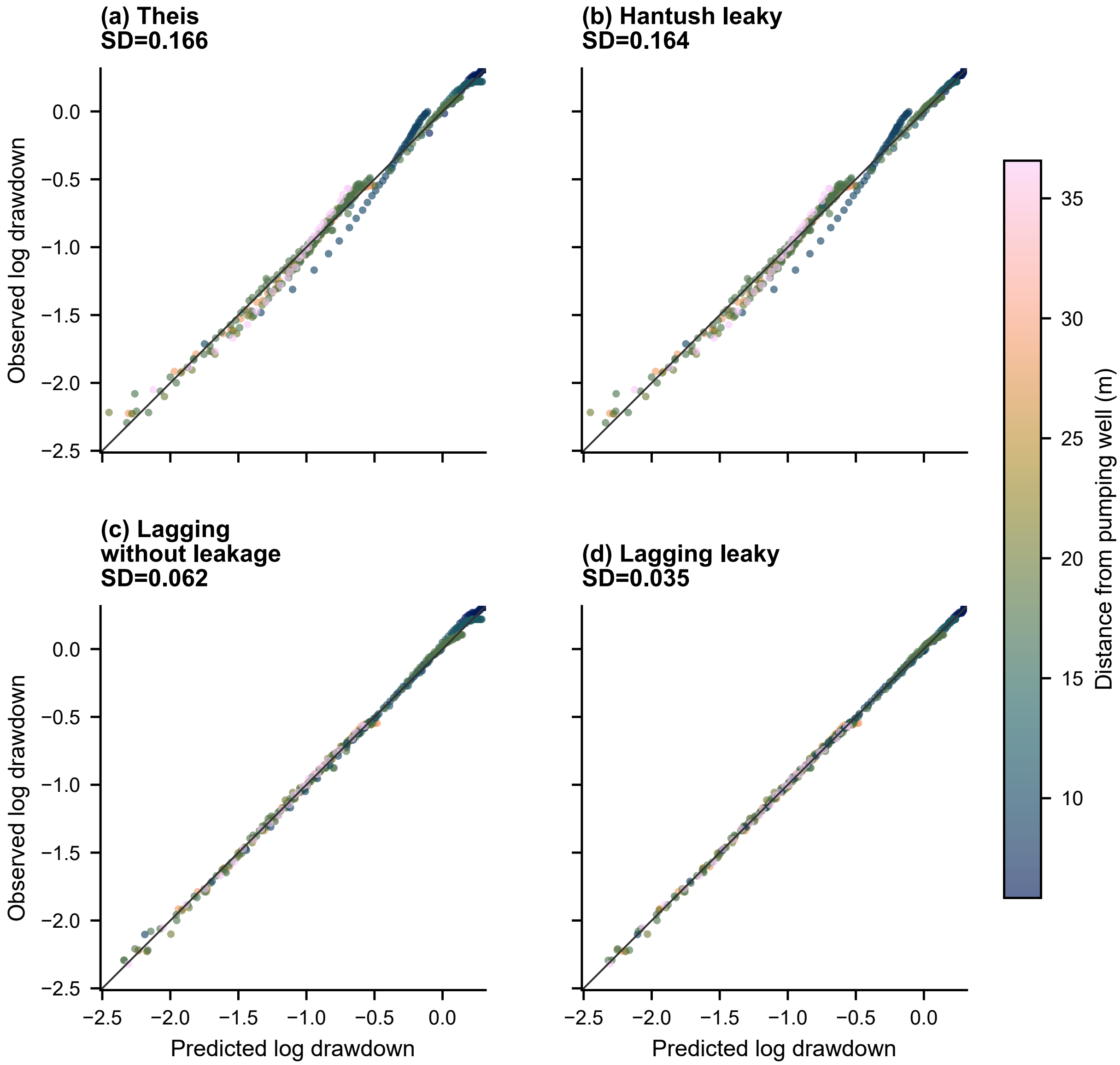

Figure S1: Massachusetts observed and predicted log drawdown comparison across (a) Theis confined; (b) Hantush–Jacob leaky; (c) lagging Darcy without leakage; and (d) lagging Darcy with leakage, with marker color for distance from the extraction well.

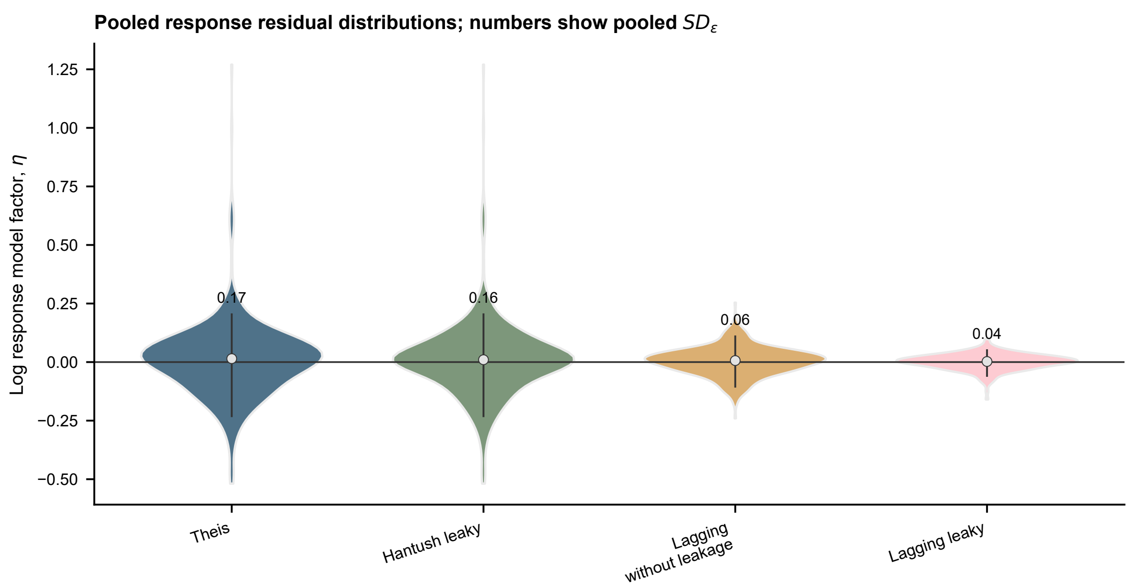

Figure S2: Massachusetts log model factor distributions across (a) Theis confined; (b) Hantush–Jacob leaky; (c) lagging Darcy without leakage; and (d) lagging Darcy with leakage, with vertical intervals for the 5th–95th percentile range, markers for medians, and numeric labels for pooled SD_ϵ.

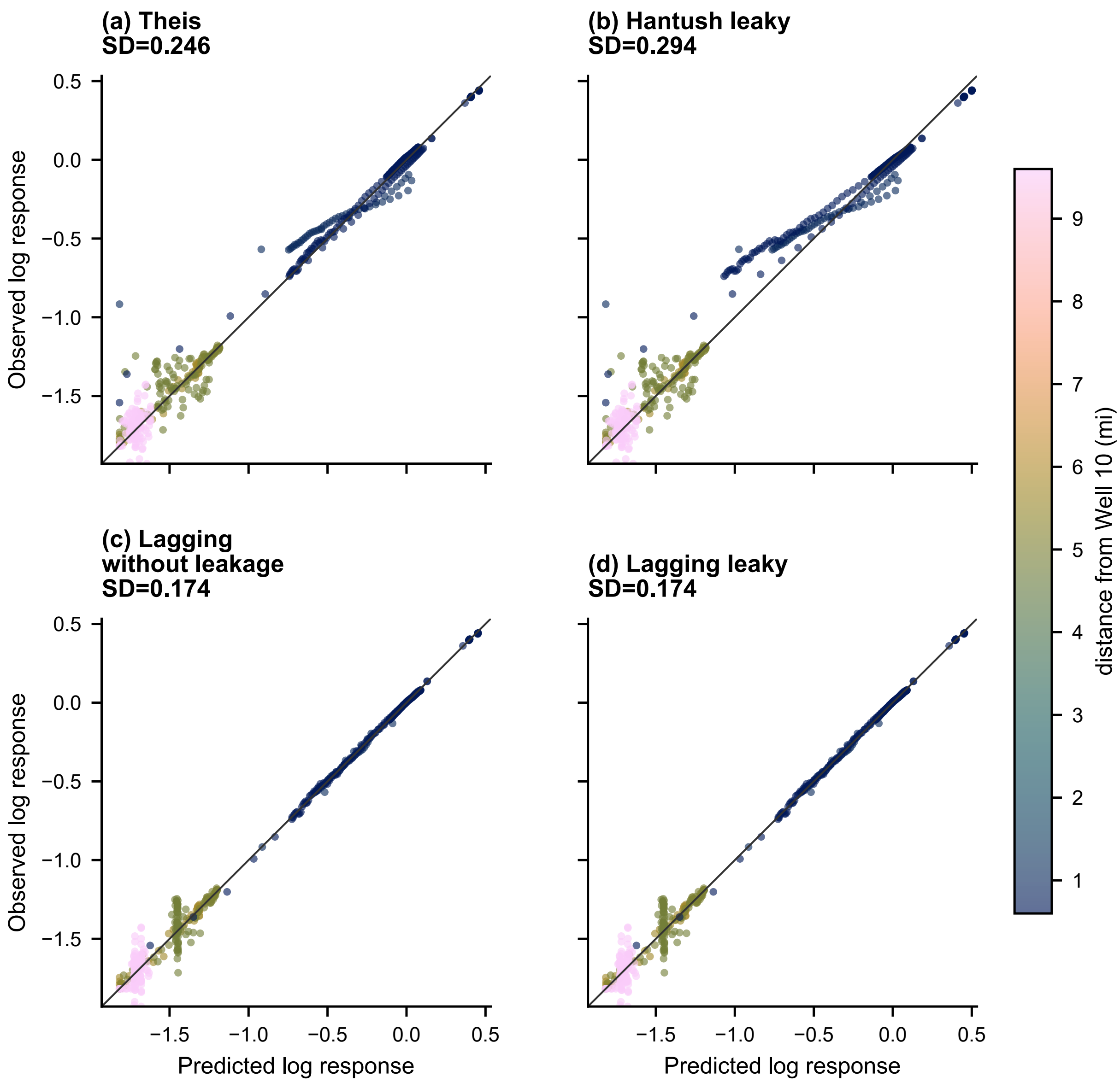

Figure S3: Lovelock Valley observed and predicted log response comparison across (a) Theis confined; (b) Hantush–Jacob leaky; (c) lagging Darcy without leakage; and (d) lagging Darcy with leakage, with marker color for distance from Well 10.

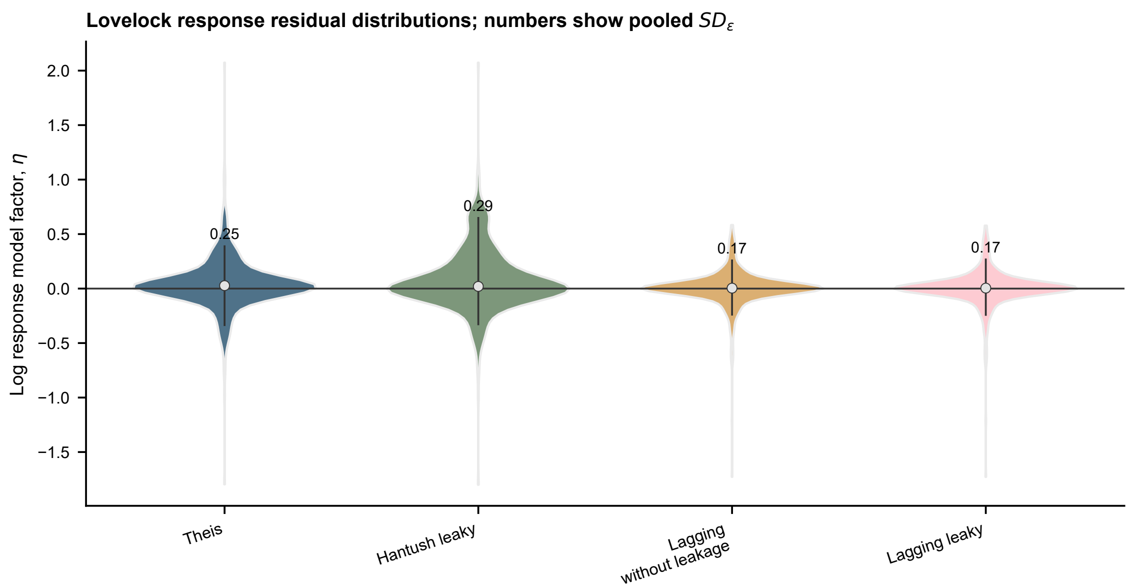

Figure S4: Lovelock Valley log model factor distributions across (a) Theis confined; (b) Hantush–Jacob leaky; (c) lagging Darcy without leakage; and (d) lagging Darcy with leakage, with vertical intervals for the 5th–95th percentile range, markers for medians, and numeric labels for pooled SD_ϵ.